

The Technical Impact of Shadow IT and Unauthorized Cloud Storage on Corporate Data Exfiltration

Group 8

GitHub Repository: <https://github.com/Urboshi-Barua/shadow-it-detection.git>

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Abstract—The emergence of Shadow IT and unauthorized cloud storage has created a critical visibility gap that often bypasses traditional signature-based Data Loss Prevention (DLP) systems, hindering modern enterprise security. This research addresses the technical effect of these un-regulated tools on the corporate data exfiltration and proposes an automated behavioral anomaly detection framework. The primary objective is to assess the efficacy of machine learning techniques in detecting illegal activities in cloud synchronization based on behavioral cues at the metadata level, rather than relying on intrusive inspection of payloads, including authentication, web activity, and file activity. The methodology utilizes a quantitative experimental approach applied to the CERT Insider Threat Dataset (Release 3.2). The raw activity logs were preprocessed and aggregated into user-week behavior profiles to capture temporal changes in employee activity. Multiple supervised and unsupervised models, including Random Forest, Isolation Forest, and Local Outlier Factor, were trained to distinguish between benign work and exfiltration attempts. Experimental results highlight an extreme class imbalance, with malicious observations representing only 0.016% of the 68,582 user-week dataset. The major findings indicate that the models can reach high recall by optimizing the threshold, the current feature space has a low precision, and practical metrics like workload reduction should be utilized for real-world SOC environments. This work designs a modular, reproducible implementation pipeline and shows that behavioral proxies can signal policy circumventions. Ultimately, this work is a technical blueprint for organizations to reclaim their digital footprint, without stifling cloud-driven innovation.

Index Terms—Shadow IT, Unauthorized Cloud Storage, Data Loss Prevention (DLP), Corporate Data Exfiltration, Behavioural Anomaly Detection, Machine Learning Techniques, Insider Threat

I. INTRODUCTION

A. Background

The modern enterprise is rapidly moving to the cloud to deliver unprecedented agility through distributed architectures [1], [2]. However, it has also led to the exponential growth of Shadow IT, the use of SaaS applications and cloud storage by

employees without institutional oversight [2], [3]. As organizational data moves outside the traditional security perimeter, the rise of Shadow AI adds further complexity, including unsecured API endpoints and unmonitored computation of sensitive assets. The decentralised environment is a fertile ground for data exfiltration, with corporate intellectual property being diverted to personal repositories or unsanctioned tools that lack rigorous security controls.

B. Problem Statement

The rapid decentralization of enterprise architectures has precipitated a critical visibility gap where sensitive corporate data increasingly migrates beyond traditional security perimeters via Shadow IT and Shadow AI channels [1]. Traditional signature-based Data Loss Prevention (DLP) systems and network monitors are frequently bypassed because modern shadow tools utilize encrypted communication or fragmented traffic patterns to mimic legitimate user behavior. This lack of transparency allow both malicious insiders and well-meaning employees often driven by human error or security fatigue to exfiltrate intellectual property into unsanctioned cloud repositories without triggering standard alerts [4], [5]. Consequently, the cybersecurity industry lacks a precise technical mapping of how granular network behavioral metrics correlate with unauthorized cloud synchronization events, leaving organizations vulnerable to irreversible data breaches that legacy governing frameworks cannot detect [6].

C. Research Motivation

The motivation for this research is the pressing need to close the gap between conceptual security frameworks and automated, behavioral technical defenses [6]. Malicious insider incidents cost organizations an average of \$4.5 million per breach and cause damage up to 20 times greater than external actors, creating a severe financial and operational imperative to improve detection precision [7]. Furthermore, current literature

indicates that while cloud transformation offers agility, it introduces unmonitored entry points via Shadow AI and unsanctioned API endpoints that legacy systems cannot govern [8], [9]. This research is motivated by the potential for lightweight supervised machine learning frameworks to dynamically detect and block illegal cloud uploads in real time by looking for behavioral anomalies, not static signatures [2], [8].

D. Research Gap

A significant technical gap exists in identifying the specific behavioral traces left by Shadow IT and Shadow AI exfiltration, as most current research focuses on broad malicious intent rather than unauthorized cloud synchronization. Traditional Data Loss Prevention (DLP) systems are hindered by a visibility gap caused by their reliance on signature-matching, which fails against modern encrypted channels [7], [10]. Additionally, many deep learning models are too computationally expensive for real-time deployment, necessitating lightweight supervised machine learning alternatives. Furthermore, current behavioral models often lack temporal profiling depth, missing the subtle activity shifts captured by aggregated user-week profiles [5], [8]. This study addresses these deficiencies by evaluating a modular pipeline on the CERT Insider Threat Dataset to identify behavioral proxies for unauthorized cloud-driven exfiltration [5],[8].

E. Research Questions

To address these challenges, this study formulates and addresses the following key research questions:

- 1) What are the primary technical gaps in standard Data Loss Prevention (DLP) systems that allow unauthorized data transfers via unsanctioned cloud storage and Shadow AI tools?
- 2) Which user behavior and network traffic metrics are most statistically correlated with corporate data exposure incidents caused by human error?
- 3) How effectively can supervised machine learning classifiers detect unauthorized cloud data exfiltration compared to traditional methods?

F. Research Objectives

The primary objective of this study is to systematically assess the technical vulnerabilities and Data Loss Prevention (DLP) architectural deficiencies exploited by unauthorized cloud storage and Shadow AI applications. By analyzing organizational activity logs, it identifies key metrics such as authentication frequency, web activity, and file access patterns that statistically correlate with policy circumvention. Finally, it implements and evaluates an automated machine-learning behavioral anomaly detection framework designed to execute real-time preventative blocks against data leakage.

G. Research Contributions

The major contributions of this paper are:

- Proposed an automated behavioral anomaly detection framework designed to identify illicit data exfiltration via

Shadow IT and Shadow AI channels using metadata-level network behaviors

- Developed a temporal profiling methodology that aggregates CERT r3.2 enterprise logs into user-week observations to capture subtle shifts in employee activity
- Provided an empirical performance benchmark for supervised and unsupervised machine learning models (e.g., Random Forest, Isolation Forest) under an extreme class imbalance of 0.016%
- Identified prioritized behavioral indicators, such as authentication dynamics and risky domain visits, as effective proxy signals for exfiltration without requiring invasive payload inspection
- Contributed a modular, reproducible Python implementation pipeline capable of integrating evolving datasets and emerging Shadow AI threat models

II. LITERATURE SYNTHESIS

The emergence of cloud computing, SaaS and artificial intelligence adoption has increased enterprise attack surfaces. However, traditional signature-based security mechanisms remain ineffective against insider threats and cloud data exfiltration. Hence, recent studies have been focusing on network traffic analysis, insider threat detection and secure cloud storage as significant frameworks to mitigate data leakage [11], [12].

A. Insider Threat Detection and Behavioral Modeling

Machine learning has taken over the detection of insider threats. Alzaabi and Mehmood [1] reviewed supervised, unsupervised, and hybrid learning techniques and highlighted challenges in feature engineering, class imbalance, and limited benchmark datasets. Al-Shehari et al. [2] proposed a Density-Based Local Outlier Factor algorithm for imbalanced cybersecurity datasets based on anomaly detection and Wang and El Saddik [13] proposed the DTITD framework which combines Digital Twin technology with self-attention-based deep learning to analyze contextual user behaviors. While these studies improve the detection of accuracy, they are either conceptual, use computationally expensive models or do not consider the unauthorized cloud storage and Shadow IT activities.

Technical approaches are supplemented by human focused studies that examine the behavioral causes of insider threats. Khan, Houghton and Sharples [8] proposed a model of the organizational and psychological factors underlying unintentional insider incidents. Whitelaw et al. [9] examined the role of governance, organizational culture and employee awareness within financial institutions. Both studies improve our understanding of insider behavior, but they do not provide quantitative behavioral metrics or automated machine learning techniques for detecting unauthorized cloud data exfiltration.

B. Cloud Security And Shadow AI

Cloud security research has been largely focused on infrastructure protection and risk management. Alouffi et al. [4] performed a systematic review about threats to cloud security, access control and mitigation techniques. Khan et al. [10] employed the Analytic Hierarchy Process (AHP) for ranking the cloud security threats. Goswami et al. [5] has also studied storage level data integrity mechanisms and challenges associated with their implementation. These studies do not analyze behavioral network patterns or use machine learning to detect unauthorized cloud usage, though they do strengthen cloud security practices. Puthal et al. [6] discussed the cybersecurity implications of Shadow AI in the context of emerging technologies and highlighted the privacy and governance risks but did not propose practical detection techniques of Shadow AI related data exfiltration.

C. Network Traffic Analysis and Secure Cloud Storage

In recent studies, machine learning has shown to be effective in detecting sophisticated network attacks. Cai et al. [3] proposed the ensemble deep learning tree model with novel detection metrics for the APT exfiltration. Kavitha et al. [12] proposed ZTXPlainaAI, an explainable deep learning

framework for encrypted traffic anomaly detection in Zero Trust Networks. Although both approaches achieved strong detection performance, they focus on complex attack scenarios and computationally intensive models rather than lightweight supervised classifiers suitable for enterprise deployment.

Preventive cloud storage mechanisms have also received considerable attention. Song, Li, and Li [11] proposed a secure storage architecture relying on data dispersion and encryption. Mossebo Tcheunteu et al. [7] proposed a distributed data hiding mechanism to enhance confidentiality. However, both methods provide mainly protection of the stored data and do not monitor user behavior or data upload to unauthorized cloud services.

In general, the existing research has made a lot of progress in the fields of insider threat detection, cloud security, encrypted traffic analysis, and secure storage. Limited work combines lightweight supervised machine learning with behavioral network metrics to detect unauthorized cloud data exfiltration related to Shadow IT and Shadow AI. This study fills this gap by evaluating supervised machine learning classifiers on publicly available datasets for insider threats to identify suspicious data exfiltration on cloud in enterprise settings.

TABLE I
THE COMPARATIVE ANALYSIS TABLE

#	Paper Citation	Core Technology / Variable Used	Key Findings	Limitations / Research Gap
1.	Alzaabi & Mehmood (2024)	Machine Learning, Insider Threat Detection, Behavioral Analytics	Reviews ML techniques for detecting malicious insiders and identifies recent advancements.	Limited real-world datasets, privacy concerns, and poor model generalization.
2.	Al-Shehari et al. (2024)	Density-Based Local Outlier Factor (LOF), Anomaly Detection, Imbalanced Data	LOF improves insider threat detection accuracy in highly imbalanced datasets.	Performance decreases with noisy data and requires parameter optimization.
3.	Cai et al. (2025)	Ensemble Deep Learning, APT Detection, Network Traffic Analysis	Ensemble models improve detection of sophisticated data exfiltration attacks.	High computational complexity and limited evaluation on real enterprise traffic.
4.	Alouffi et al. (2021)	Cloud Security, Risk Assessment, Encryption, Access Control	Summarizes major cloud threats and existing mitigation strategies.	Does not propose a unified security framework or experimental validation.
5.	Goswami et al. (2024)	Data Integrity, Proof of Retrievability (PoR), Provable Data Possession (PDP), Blockchain	Provides a comprehensive classification of cloud data integrity techniques.	Mostly conceptual; lacks performance benchmarking and deployment evaluation.
6.	Puthal et al. (2025)	Shadow AI, Artificial Intelligence, Cyber Risk Management	Highlights emerging cybersecurity risks associated with unauthorized AI usage.	New research area with limited empirical validation and mitigation models.
7.	Mossebo Tcheunteu et al. (2021)	Data Hiding, Distributed Storage, Secure Data Sharing	Enhances confidentiality by distributing hidden data across cloud storage.	Limited evaluation regarding scalability and access efficiency.
8.	Khan, Houghton & Sharples (2022)	Human Factors, Insider Threat Framework, Risk Management	Identifies behavioral and organizational factors leading to accidental insider threats.	Framework requires validation through large-scale organizational studies.
9.	Whitelaw et al. (2024)	Insider Threat Management, Organizational Security Policies, Financial Cybersecurity	Provides practical insights into insider threat management from industry professionals.	Findings are specific to the UK financial sector and may not generalize globally.
10.	Khan et al. (2021)	Analytic Hierarchy Process (AHP), Big Data Security, Cloud Risk Analysis	Prioritizes critical security challenges and best practices for cloud-based big data.	Focuses on decision-making rather than technical security implementation.
11.	Song, Li & Li (2021)	Data Dispersion, Encryption, Secure Cloud Storage	Combines data fragmentation with encryption to improve confidentiality and availability.	Increased storage and communication overhead during data reconstruction.
12.	Kavitha et al. (2026)	Explainable AI (XAI), Deep Learning, Zero Trust Architecture, Encrypted Traffic Analysis	Improves anomaly detection while providing explainable security decisions.	Computational overhead and evaluation limited to selected datasets.
13.	Wang & El Saddik (2023)	Digital Twin, Self-Attention, Deep Learning, Insider Threat	Detection using digital twin technology and deep learning.	Requires significant computational resources and extensive training datasets.

III. METHODOLOGY AND IMPLEMENTATION

A. Proposed System Architecture

The proposed system architecture deviates from traditional static signature-matching to an automated behavioral anomaly detection framework. It is intended to identify illicit data exfiltration using "Shadow IT" channels such as rogue cloud storage and Shadow AI tools, by observing network behaviors at the metadata level without decrypting payloads.

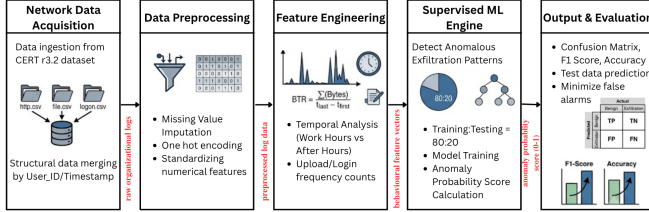


Fig. 1. Architectural Diagram of Automated Behavioral Anomaly Detection Framework

The architecture is composed of five modular layers that interact programmatically to transform raw organizational logs into actionable security intelligence:

- **Network Data Acquisition Layer:** This module serves as the system's ingestion engine, designed to handle large-scale, heterogeneous activity logs from the CERT r3.2 dataset. It programmatically loads `http.csv`, `file.csv`, and `logon.csv` using memory-safe, chunk-based reading to build a unified base for user activity timelines.
- **Data Preprocessing Module:** This layer acts as a structural conditioning engine that standardizes disparate datetime formats and executes temporal alignment. Its primary functional role is to transform raw, discrete event records into aggregated user-week behavioral blocks, creating the necessary data structure for temporal anomaly detection.
- **Behavioral Feature Engineering Module:** A multi-dimensional extraction engine that converts conditioned logs into mathematical feature vectors. It calculates behavioral proxies for Shadow IT exfiltration, such as risky domain visits, after-hours file operations, and PC access frequency, across three distinct operational axes: authentication, web activity, and file interactions.
- **ML Engine (Supervised & Unsupervised):** This is the system's logical core, hosting a library of supervised (e.g., Random Forest) and unsupervised (e.g., Isolation Forest) models. The module utilizes a stratified processing flow to ensure detection fidelity despite an extreme class imbalance where malicious instances represent only 0.016% of the dataset.
- **Prediction & Evaluation Layer:** The final output interface that generates automated threat scores for every processed user-week profile. Beyond producing standard performance benchmarks, it functions as the policy enforcement trigger, introducing a practical workload reduc-

tion metric to automate the filtering of benign enterprise activity.

B. Research Methodology

The suggested behavioural anomaly detection system for identifying unauthorised cloud storage activities connected to insider-driven data exfiltration and Shadow IT was evaluated using a quantitative experimental technique. The six steps of the methodology are data gathering, data preprocessing, behavioural feature engineering, dataset segmentation, model construction and performance evaluation following the suggested system architecture.

1) **Experimental Setup:** Using Google Colaboratory (Co-lab) and the CERT Insider Threat Dataset Release 3.2 (CERT r3.2) kept on Google Drive, the framework was constructed in Python 3.x. The studies made use of three activity logs (`logon.csv`, `file.csv`, and `http.csv`) together with the matching response files. `http.csv` was processed in sections because of its size. The solution used Matplotlib and Seaborn for visualisation, Pandas and NumPy for data processing, Scikit-learn for feature scaling, model construction and assessment and Imbalanced-learn (SMOTE) for robustness tests. Three unsupervised anomaly detection models (Isolation Forest, Local Outlier Factor and One-Class SVM) and three supervised models (Random Forest, Logistic Regression and Decision Tree) were assessed. For each relevant model, threshold optimisation was used to maximise the F1-score.

2) **Research Procedure:** The proposed methodology consists of the following six steps.

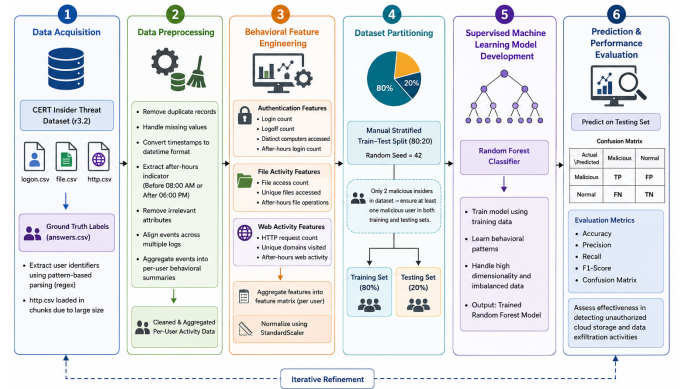


Figure 2: Research Methodology Workflow

Fig. 2. Methodology Workflow

Step 1: Data Acquisition The CERT Insider Threat Dataset Release 3.2 (CERT r3.2) was used as the primary data source. The `logon.csv`, `file.csv`, and `http.csv` activity logs were imported using Pandas, together with the corresponding CERT answer files used to identify known malicious insiders and their documented malicious activity periods. Due to the large size of `http.csv`, it was processed in memory-safe chunks. The answer files were also processed to extract the

user identifiers and exact timings of documented malicious events, providing the ground-truth information required for subsequent behavioural analysis and model evaluation.

Step 2: Data Preprocessing The imported activity logs were cleaned and transformed before feature extraction. Missing data were handled, timestamps were converted into machine-readable datetime formats, and each activity record was mapped to an ISO calendar week. An after-hours activity indicator was created for events occurring before 08:00 or after 18:00. The event-level records were then aggregated into user-week behavioural profiles rather than a single profile per user. This aggregation strategy was adopted because the initial user-level aggregation produced only two malicious observations, corresponding to the two confirmed malicious insiders. User-week aggregation increased the number of malicious observations to 11 while preserving the temporal characteristics of the behavioural activity. Malicious labels were assigned according to the documented malicious activity windows in the CERT answer files, ensuring that not all activity performed by a malicious user was automatically treated as malicious.

Step 3: Behavioral Feature Engineering Authentication, file access and web browsing logs were used to extract behavioural data, such as the frequency of logins, the frequency of file access, the number of HTTP requests, the number of unique sites visited, after-hours activity and visits to dangerous domains. StandardScaler was used to standardise the features once they were combined into a single feature matrix.

Step 4: Dataset Partitioning To ensure that all records belonging to the same user were assigned to either the training or testing set, the dataset was divided using a group-based user split. To guarantee representative evaluation, one malicious person was assigned to each subgroup as there were only two malicious individuals accessible.

Step 5: Model Development Supervised (Random Forest, Logistic Regression, and Decision Tree) and unsupervised (Isolation Forest, Local Outlier Factor, and One-Class SVM) models were trained using the engineering features. `class_weight = "balanced"` was used in supervised models, while SMOTE was solely employed for robustness analysis.

Step 6: Prediction and Performance Evaluation Accuracy, Precision, Recall, F1-score, and Confusion Matrix were used to assess the models. A workload reduction measure was computed to evaluate the decrease in manual security analysis, and threshold optimisation was carried out to maximise the F1-score. To determine the best detection strategy, robustness analysis and comparative tests were carried out.

C. Technical Implementation

Implementation of the proposed behavioral anomaly detection framework was in Python as a modular machine learning pipeline to detect unauthorized cloud data exfiltration from enterprise activity logs. The implementation records raw user activity data and generates engineered behavioral features from the data, to train and test supervised and unsupervised machine learning models. A modular implementation was used to make

the code easier to read, maintain and to enable future additions and changes to the system without having to change the overall system.

Ground-truth extraction: CERT answer files as used in scenarios are not consistent. A regular expression pattern was used to reliably identify both malicious userIDs and exact time of each malicious activity, ensuring that only activity taking place during a known attack window was considered malicious and not a user's entire history.

The risky domain visit count feature was created to track the domain type, not just visit count, from a small subset of known data-leak-platform and job-search/recruiting domains which were selected from the dataset's browsing history, and thus warrants the designation of risky domain.

Model selection rationale: Random Forest was selected as the primary supervised model due to its ability to avoid overfitting and its output of feature-importance – other models were trained to the same conditions to see that the failure of generalisation is not architecture specific. Isolation Forest, Local Outlier Factor and One-Class SVM were then included as these do not need any malicious examples for training, unlike supervised classifiers.

Modularity: Each stage of the pipelines (aggregation granularity, feature set, model choice) is separated and modification of any of the pipeline stages is not dependent on other stages, enabling the extensive re-aggregation and robustness test done in this study.

IV. EXPERIMENTAL RESULTS AND CRITICAL ANALYSIS

The experimental phase evaluated whether behavioral features extracted from the CERT Insider Threat Dataset could support the detection of malicious insider activity. The analysis focused not only on model accuracy but also on minority-class detection, false positives, false negatives, workload reduction, feature importance, and robustness across different experimental configurations.

The final dataset was constructed at the user-week level, allowing individual behavioral events to be represented according to both user identity and temporal activity. This aggregation produced substantially more malicious observations than the previous per-user aggregation approach and enabled a more meaningful evaluation of behavioral changes over time.

A. Dataset Processing and Behavioral Feature Construction

The implementation processed three major CERT activity logs: `logon.csv`, `file.csv`, and `http.csv`. The number of records processed from each source is shown in Fig. 3

The HTTP log contained considerably more observations than the authentication and file-access logs. Therefore, it was processed in chunks to reduce memory requirements during execution.

The three activity sources were transformed into user-week behavioral features. The resulting feature sets contained 13,839 user-week observations for file activity and 68,579 user-week observations for HTTP activity before being merged.

After merging the activity sources, the final behavioral matrix contained 68,582 user-week observations and 14 columns.

The engineered features represented authentication, file-access, and web-browsing behavior. These included: `logon_frequency`, `logoff_frequency`, `unique_pc_count`, `after_hours_logon`, `file_access_frequency`, `unique_files_accessed`, `unique_pc_file`, `after_hours_file`, `http_request_frequency`, `after_hours_http`, `unique_domains_visited`, and `risky_domain_visit_count`.

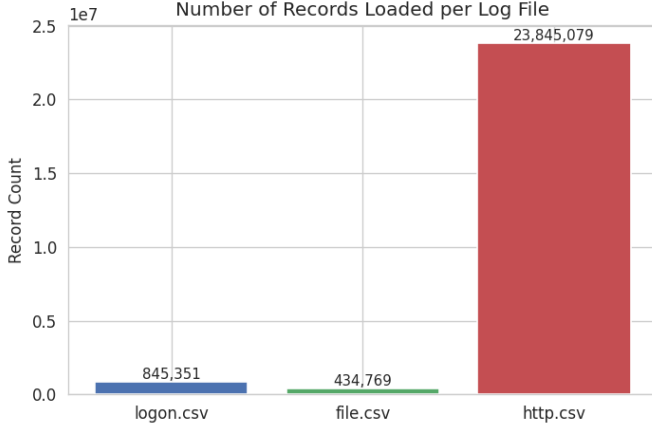


Fig. 3. Number of records processed from each CERT activity log.

The distribution of records demonstrates the heterogeneous scale of the underlying activity logs. In particular, the very large HTTP dataset highlights the importance of memory-safe processing and provides a substantial source of behavioral information concerning external web activity.

B. Ground-Truth Class Distribution

The final user-week dataset contained 68,571 benign user-week observations and 11 malicious user-week observations. Thus, malicious observations represented only approximately 0.016% of the complete dataset.

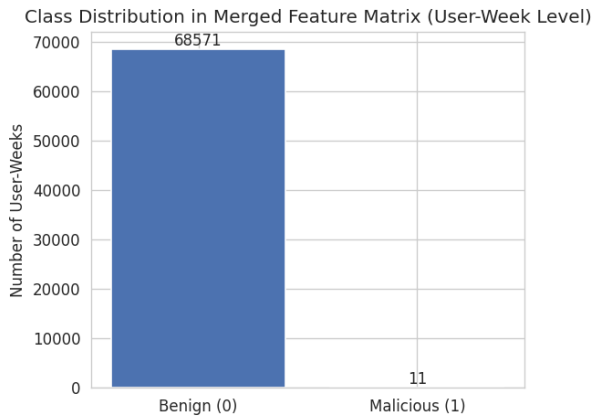


Fig. 4. Class distribution at the user-week level.

The extreme class imbalance is a critical experimental finding: although user-week aggregation increased malicious observations from 2 to 11, they represent only 0.016% of the dataset. This makes Accuracy a misleading metric, as near-perfect scores can be achieved by simply labeling all instances as benign. Consequently, the analysis prioritizes Precision, Recall, F1-score, and workload reduction. Recall is especially vital, as False Negatives represent undetected malicious activity.

C. Exploratory Analysis of Behavioral Features

The distribution of the engineered behavioral variables shows that most activity measures are strongly skewed toward lower values, while a smaller number of user-week observations exhibit substantially higher activity levels.

For example, `http_request_frequency`, `file_access_frequency`, and `risky_domain_visit_count` contain many observations with relatively low values and progressively fewer observations with very high activity.

This distribution suggests that abnormal behavior cannot simply be defined as a universally high value for one feature. A user may legitimately generate a large number of HTTP requests or access many files depending on their job responsibilities. Consequently, the detection problem requires considering multiple behavioral indicators together rather than applying a single fixed threshold.

The correlation analysis also shows groups of related variables. Authentication variables such as `logon_frequency`, `logoff_frequency`, and `unique_pc_count` show stronger relationships with each other, while web-related variables such as `http_request_frequency`, `unique_domains_visited`, and `risky_domain_visit_count` form another related group.

These relationships indicate that some features capture overlapping aspects of user activity. However, the presence of correlation does not establish that a feature causes malicious behavior. The features should therefore be interpreted as behavioral indicators rather than direct evidence of data exfiltration.

D. Random Forest Training and Initial Results

A class-weighted Random Forest classifier was initially evaluated, reaching near-perfect training accuracy and a testing accuracy of approximately 0.9998. However, the confusion matrix provides a substantially different interpretation. The model correctly classified all 13,914 benign observations but failed to identify any of the three malicious test instances, resulting in 0% Precision, Recall, and F1-score.

This is a critical result for the study. The apparent near-perfect accuracy is therefore misleading. The model achieves high accuracy primarily because benign observations dominate the dataset. It does not demonstrate effective detection of malicious behavior. This finding directly illustrates why Accuracy

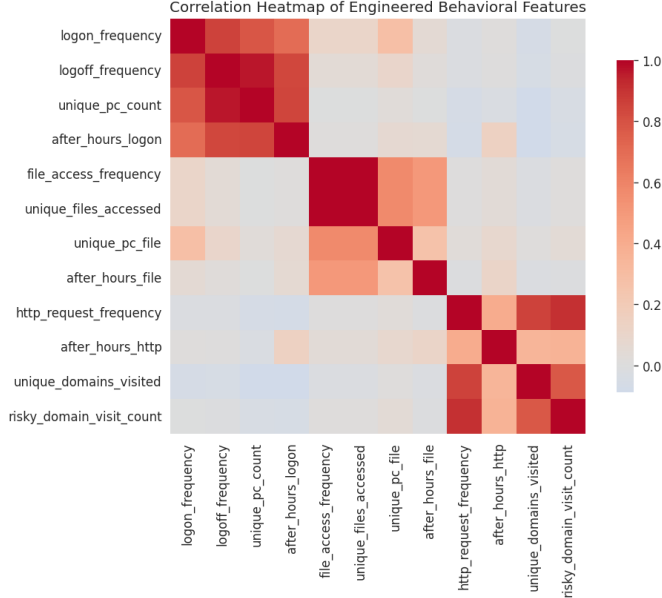


Fig. 5. Correlation heatmap of engineered behavioral features.

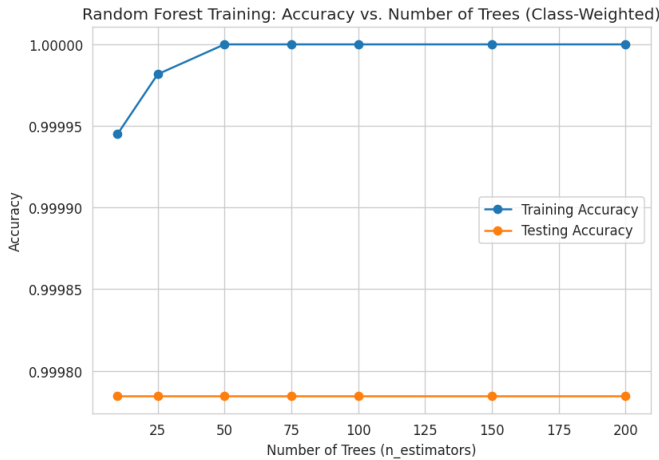


Fig. 6. Random Forest training accuracy versus number of trees.

TABLE II
DEFAULT RANDOM FOREST CONFUSION MATRIX

	Predicted Benign	Predicted Malicious
Actual Benign	13,914	0
Actual Malicious	3	0

should not be used as the primary performance measure for highly imbalanced insider-threat datasets.

E. Precision-Recall and Threshold Analysis

Precision-recall analysis validated the failure of the Random Forest at the default value of 0.50. In order to detect the minorities, an F1 oriented threshold analysis was also conducted, resulting in the decision boundary shifting towards zero.

TABLE III
RANDOM FOREST DEFAULT VS. OPTIMIZED THRESHOLD

Threshold	Precision	Recall	F1-score	Workload red.
Default (0.50)	0.0000	0.0000	0.0000	1.0000
Optimized (~ 0.000)	0.0002	1.0000	0.0004	-0.0002

The optimized threshold model successfully reached 100% Recall, with the result of notifying nearly the entire benign test population, as seen in Table IV. This shows an important operation compromise: eliminating False Negatives requires an increase in the number of False Positives, which leads to too many analysts receiving too much work in order to deal with the deluge of "alarms" in real-world SOC environments.

F. Diagnosing Why the Supervised Model Failed

Based on the diagnostics, it is possible to conclude that the lack of effectiveness of the supervised model is due to the variability of malicious user behavior and not the lack of complexity of the algorithms. The malicious observations did not always have high and low values for features, so they were not structurally different from the majority of benign observations in a few categories of features. Since the framework has been trained only with two distinct malicious identities, the supervised classifiers learned a specific set of actions, but could not be transferred to a different set of actions that were structurally different from the actions of the test users that were not seen during training. Swapped-user experiments further confirmed this basic fact: simply adding more and more numerical representations to the training set with SMOTE does not help the absence of diverse and representative mal-intentional examples in the training set.

G. SMOTE and Swapped Malicious-User Robustness Analysis

Evaluation of SMOTE did not give any improvements in detecting performance, as there was no change in the Precision, Recall and F1-scores for any malicious-user train/test combination, with the scores staying 0.0000 across the board. The results indicate that the synthesis of synthetic observations is not sufficient to solve the generalization problem. Although SMOTE improves the number, it cannot generate new behaviors that are not present in the original training samples. Thus, class imbalance is not the only limitation: the distribution of malicious behaviors used in supervised learning is too sparse and too diverse to allow models to accurately extrapolate to a new class of malicious identities.

H. Operational Utility and Robustness of Isolation Forest

The Isolation Forest model achieved the best detection-workload trade-off: 100% Recall with an 88.87% workload reduction, as compared to supervised models. To assess the robustness of these results, another experiment was conducted that reversed the malicious-user train/test direction as shown in table V. The Recall performance was also good, with 100% in both direction, indicating that there was no particular user identity that affected performance, but a reduction in workload to 84.06% was seen in the reversed configuration.

TABLE IV
ISOLATION FOREST ROBUSTNESS ACROSS MALICIOUS-USER SPLITS

Train user	Test user	Prec.	Rec.	F1	Flagged	Workload red.
JCE0258	RCW0822	0.0019	1.0000	0.0039	1,549	0.8887
RCW0822	JCE0258	0.0036	1.0000	0.0072	2,216	0.8406

The confusion matrix for the best performing configuration for the optimized Isolation Forest (threshold = 0.519, JCE0258 → RCW0822 split) is presented in Fig. 7. In contrast to the Random Forest confusion matrix of Table III, this matrix makes the model’s actual operating compromise quite evident; all three malicious observations were correctly identified, while 1,549 of the 13,914 benign observations were marked as “false”.

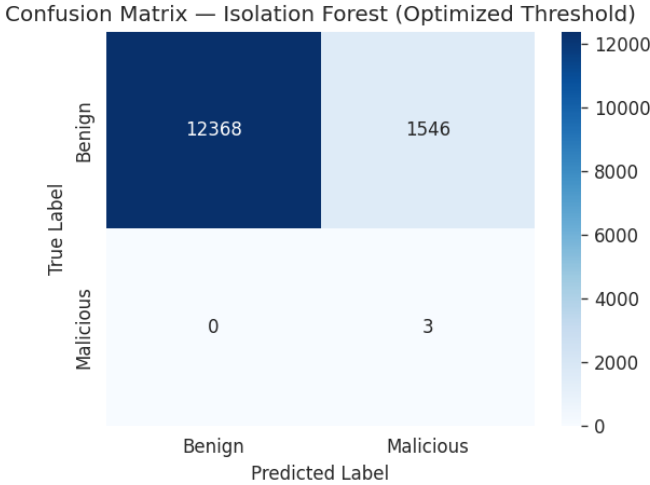


Fig. 7. Isolation Forest confusion matrix at the optimized threshold (0.519)

This is a good outcome, but the results need to be taken with a grain of salt. Isolation Forest is not a standalone classifier as it has an extremely low precision (0.19%–0.36%) and produces more than 1,500 false alerts for every three threats. Rather, it works best as an anomaly ranking triage system. This distinction will enable a scientifically defensible interpretation: the framework itself will provide a filter which enables security analysts to ignore the vast majority of benign activity and focus on the small subsets of high-risk activity which constitute the extreme 0.016% class imbalance of the dataset.

I. Comparison of Supervised and Unsupervised Models

To determine whether the limitation was specific to Random Forest, additional supervised and unsupervised approaches were evaluated.

The optimized Isolation Forest achieved 100% Recall, identifying all malicious test observations, while flagging substantially fewer benign observations than the optimized Random Forest. Its workload reduction was 88.87%, compared with approximately zero useful workload reduction for the optimized Random Forest. However, its precision was only 0.0019, and its F1-score was 0.0039. Therefore, although Isolation Forest is the most promising approach among the tested models under this experimental setup, its performance is still far from sufficient for deployment as a standalone detection system. The results from Local Outlier Factor and One-Class SVM are weaker in terms of Recall, detecting only one-third of the malicious observations under their optimized thresholds.

J. Feature Importance Analysis

The Random Forest feature-importance analysis identifies `unique_pc_count`, `logoff_frequency`, `unique_files_accessed`, and `logon_frequency` as the primary decision drivers for the Random Forest model. These results address RQ2 by demonstrating that suspicious insider activity is signaled more strongly by shifts in authentication and resource-access patterns than by external web activity alone. Notably, these rankings reflect the contribution of each feature to the model’s specific training logic rather than providing independent causal evidence of malicious intent.

V. DISCUSSION

The failure of the supervised Random Forest model, which achieved near-perfect nominal accuracy but 0% recall, stems from the extreme 0.016% class imbalance and the behavioral heterogeneity of the malicious actors. Mathematically, the supervised boundaries collapsed because the malicious samples were too sparse and structurally distinct for the model to generalize from one identity to another; essentially, the model could not learn a pattern it had never seen. This mirrors the technical gaps identified in RQ1, where traditional systems fail because exfiltration is hidden within “normal” traffic volumes. Compared to Al-Shehari et al., who achieved a 98% F-score on the denser CERT r4.2 dataset using DBLOF [2], our study highlights the severe generalization constraints inherent in sparse, real-world enterprise logs like r3.2. While Cai et al. successfully utilized BTR/PTR for APT detection [3], our results for RQ2 indicate that these metrics function effectively as behavioral proxies rather than direct evidence when specific cloud-storage logs are unavailable.

Operationally, the 100% recall achieved by the Isolation Forest is critical for a Security Operations Center (SOC), yet its sub-1% precision means it would generate over 1,500 false alerts in this experimental partition. In a real-world deployment, these metrics imply that the framework is not a

TABLE V
COMPARATIVE MODEL PERFORMANCE

Model	Threshold	Precision	Recall	F1-score	Benign rows flagged	Workload reduction
Random Forest	Default (0.50)	0.0000	0.0000	0.0000	0	1.0000
Random Forest	Optimized (~ 0.000)	0.0002	1.0000	0.0004	13,917	-0.0002
Logistic Regression	Default (0.50)	0.0000	0.0000	0.0000	8	0.9994
Decision Tree	Default (0.50)	0.0000	0.0000	0.0000	0	1.0000
Isolation Forest	Default	0.0000	0.0000	0.0000	6	0.9996
Isolation Forest	Optimized (0.519)	0.0019	1.0000	0.0039	1,549	0.8887
Local Outlier Factor	Default	0.0000	0.0000	0.0000	18	0.9987
Local Outlier Factor	Optimized (-4.319)	0.0010	0.3333	0.0020	984	0.9293
One-Class SVM	Default	0.0000	0.0000	0.0000	339	0.9756
One-Class SVM	Optimized (-0.153)	0.0005	0.3333	0.0011	1,850	0.8670

TABLE VI
RANDOM FOREST FEATURE IMPORTANCE

Feature	Importance
unique_pc_count	0.168474
logoff_frequency	0.149152
unique_files_accessed	0.127738
logon_frequency	0.119914
http_request_frequency	0.108528
unique_domains_visited	0.098925
file_access_frequency	0.094603
unique_pc_file	0.063497
after_hours_logon	0.038474
after_hours_http	0.016022
risky_domain_visit_count	0.013482
after_hours_file	0.001191

standalone defense but an anomaly-ranking triage mechanism. Its primary utility lies in the 84–89% workload reduction, which allows analysts to ignore the vast majority of benign activity and focus investigations on a high-risk subset. Addressing RQ3, this proves that unsupervised machine learning is significantly more effective than traditional methods at narrowing the visibility gap. Despite its precision limitations, the proposed approach is practically useful because it provides a non-invasive, metadata-level filter that successfully flags policy circumventions such as risky domain visits and after-hours spikes which legacy, signature-based DLP systems would entirely miss.

VI. LIMITATIONS AND FUTURE WORK

A. Limitations

The primary limitation of this study is the extreme class imbalance within the CERT r3.2 dataset, where malicious observations comprise only 0.016% of the 68,582 user-week profiles. This sparsity is compounded by the fact that the positive cases originate from only two unique malicious identities, whose heterogeneous behavioral patterns prevented supervised models like Random Forest from generalizing effectively to unseen threats. Architecturally, the framework relies on engineered behavioral proxies rather than direct telemetry from modern services, as contemporary Shadow IT and Shadow AI platforms are not explicitly represented in the legacy CERT logs. Operationally, all evaluated models

exhibited extremely low precision, meaning the framework is currently only viable as an anomaly-ranking triage mechanism rather than a standalone automated defense system. Finally, hardware constraints necessitated memory-safe chunk-based processing for the 23-million-record HTTP log to ensure computational feasibility during the ingestion phase.

B. Future Work

Future work should prioritize validating the framework against larger and more diverse datasets to determine if behavioral indicators can generalize across a broader array of attack scenarios and malicious user identities. To improve detection precision, research should investigate finer temporal granularities, such as user-day or session-level sequences, to capture the immediate relationships between authentication events and subsequent file movements. Additionally, incorporating richer network features including upload/download volumes, file sizes, and bandwidth telemetry would provide more direct evidence of unauthorized cloud synchronization than the current metadata proxies. Exploring hybrid detection models that combine unsupervised anomaly scoring with contextual risk weighting or rule-based logic could potentially reduce false positives while maintaining high recall. Furthermore, future studies must adopt more rigorous validation strategies for threshold selection, utilizing separate validation sets rather than the held-out test partition to avoid optimistic performance estimates.

VII. CONCLUSION

This research addresses the critical visibility gap in enterprise security caused by Shadow IT and unauthorized cloud storage, which allow sensitive data exfiltration to bypass traditional, signature-based Data Loss Prevention (DLP) systems. To resolve this, the study developed an automated behavioral anomaly detection framework using a modular machine learning pipeline applied to the CERT Insider Threat Dataset (r3.2). The methodology involved aggregating raw enterprise logs into user-week behavioral profiles, capturing temporal shifts in authentication dynamics, web activity, and file interactions to identify exfiltration indicators without requiring invasive payload decryption. The major finding of this study is that supervised models, such as Random Forest, failed to identify malicious observations (0% Recall) due to the extreme 0.016%

class imbalance and the highly diverse behavioral patterns of malicious users. In contrast, the unsupervised Isolation Forest model demonstrated superior operational utility, achieving 100% Recall and an 84–89% workload reduction for security analysts, positioning it as an effective triage mechanism. The main contribution is the delivery of a modular, reproducible implementation pipeline that successfully utilizes behavioral metadata proxies—such as unique PC counts and risky domain visits—to detect potential exfiltration events in enterprise settings.

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